

CAP WHITTLE, QC, Canada

WMO: 713740

Lat: 50.177N

Lon: 60.121W

Elev: 23

StdP: 14.68

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.1	-3.0	-17.9	2.1	-2.1	-12.1	2.9	-1.9	49.4	22.0	45.6	23.4	17.8	300	0.978

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.9	62.2	58.4	61.0	57.8	59.8	57.2	60.0	60.8	58.7	59.6	57.3	58.4	14.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.5	76.0	60.2	58.2	72.5	59.2	56.7	68.5	58.0	26.4	61.1	25.6	59.7	24.7	58.4	64.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
42.7	38.2	34.5	-12.9	67.1	4.1	2.6	-15.8	68.9	-18.3	70.4	-20.6	71.9	-23.6	73.7		
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.9	15.4	15.6	22.4	30.8	38.6	46.6	53.2	56.3	50.9	43.0	33.9	22.8
	DBStd	15.25	10.06	9.38	7.94	4.19	3.76	3.72	3.24	3.46	3.98	4.17	6.22	8.79
	HDD50	5524	1074	962	854	576	354	113	8	4	35	217	484	843
	HDD65	10615	1539	1382	1319	1026	819	552	365	269	422	680	934	1308
	CDD50	384	0	0	0	0	0	11	108	200	64	1	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	16.9	19.6	19.1	18.7	16.6	14.7	13.9	13.1	13.5	16.3	17.5	19.7	20.5
Precipitation	PrecAvg	38.9	3.0	2.5	2.7	2.6	3.0	3.1	4.0	3.9	3.6	3.6	3.6	3.3
	PrecMax	48.4	4.9	4.4	5.2	3.7	6.3	5.5	7.0	6.7	5.9	6.1	5.7	5.6
	PrecMin	31.4	1.2	1.1	1.3	1.5	1.1	1.3	2.4	1.4	1.7	1.7	1.7	1.8
	PrecStd	3.5	0.9	0.8	0.9	0.7	1.2	0.9	1.1	1.3	1.0	1.0	1.0	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.8	36.5	37.9	40.6	51.5	58.6	63.7	65.1	61.4	53.8	47.6	41.0
		MCWB	36.5	36.0	36.1	35.5	46.5	50.9	57.2	59.6	56.3	51.7	46.8	40.0
	2%	DB	34.6	33.5	35.3	38.3	47.7	55.4	61.1	63.1	59.5	51.2	45.4	38.5
		MCWB	33.7	32.6	33.3	35.1	43.9	50.1	56.2	59.4	55.3	49.0	44.1	37.4
	5%	DB	32.1	31.0	33.3	36.8	45.5	53.3	59.4	61.8	58.0	50.0	43.5	36.7
		MCWB	30.8	30.4	31.6	34.3	42.4	49.6	55.9	59.0	54.5	48.1	42.1	35.5
	10%	DB	29.8	29.1	31.5	35.4	43.6	51.4	57.7	60.8	56.5	48.8	41.8	34.7
		MCWB	28.5	28.1	30.3	33.5	41.2	49.1	55.2	58.5	53.6	46.9	40.3	33.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.1	36.4	37.5	37.8	48.3	53.1	60.2	61.9	57.4	52.4	46.7	40.2
		MCDB	37.5	36.5	37.8	39.4	50.5	54.7	61.6	62.5	58.1	53.2	47.0	40.6
	2%	WB	34.2	32.4	34.1	35.9	45.4	51.6	58.4	60.8	55.6	49.9	44.1	37.9
		MCDB	35.0	32.6	35.0	37.0	47.5	52.9	59.5	61.5	56.8	50.7	44.7	38.7
	5%	WB	30.8	29.9	32.2	34.8	43.4	50.7	57.0	59.7	54.3	48.6	42.2	36.2
		MCDB	31.5	30.4	33.3	35.9	45.3	52.0	58.1	60.5	55.8	49.7	43.0	37.0
	10%	WB	28.2	26.9	30.7	33.8	41.8	49.6	55.8	58.8	53.2	47.3	40.5	33.9
		MCDB	29.0	27.8	31.6	34.9	43.6	51.1	57.0	59.6	54.8	48.4	41.6	35.0
Mean Daily Temperature Range	5% DB	MDBR	12.2	11.6	10.2	6.1	5.9	6.2	6.0	5.9	6.0	5.6	7.0	10.0
		MCDBR	12.6	10.5	9.9	6.9	8.9	8.9	8.2	7.0	6.8	6.2	7.3	9.8
		MCWBR	12.7	12.3	10.7	5.6	5.9	6.0	5.4	5.8	6.6	6.0	8.1	9.6
	5% WB	MCDBR	12.4	12.3	10.3	5.9	7.4	7.1	6.5	6.4	6.0	5.7	7.4	9.5
		MCWBR	12.8	12.7	10.3	5.2	5.7	5.5	5.3	5.3	6.7	5.8	8.5	10.2
Clear-Sky Solar Irradiance	taub		0.282	0.292	0.304	0.338	0.362	0.380	0.412	0.384	0.354	0.334	0.300	0.282
	taud		2.388	2.342	2.367	2.396	2.426	2.394	2.357	2.435	2.492	2.501	2.513	2.415
	Ebn at Noon		243	270	287	288	284	279	267	270	267	252	237	226
	Edh at Noon		20	27	31	34	35	36	37	33	28	23	18	17
All-Sky Solar Radiation	RadAvg		307	567	990	1374	1622	1703	1627	1430	1071	667	340	235
	RadStd		31	57	102	148	117	167	150	80	73	46	31	25

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	N/A	+2.75	N/A	+0.74	+0.77	N/A	N/A	N/A	N/A

Nomenclature: See separate page